Lecture #17: The Practice of Confidence Intervals

STAT 200 – Introductory Statistics

Torin Quinlivan

Knox College

October 28, 2026

One-Parameter Procedures: μ

Parametric Procedure: Student's t-test

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Data generated from Normal distribution
  - Requirement test: Shapiro-Wilk test
    - `shapiroTest(x)`
- `R` function: `t.test(x)`

One-Parameter Procedures: μ (and $\tilde{\mu}$)

Non-Parametric Procedure #1: Wilcoxon procedure

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Data generated from symmetric distribution
  - Requirement test: Hildebrand Rule
    - `hildebrand.rule(x)`
- `R` function: `wilcox.test(x, conf.int=TRUE)`

One-Parameter Procedures: μ

Non-Parametric Procedure #2: Non-parametric bootstrap

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Nothing!
- `R` code:

```
st = numeric()

for (i in 1:1e4) {
    x = sample(y, replace=TRUE)
    st[i] = mean(x)
}

quantile(st, c(0.025, 0.975))
```

One-Parameter Procedures: p

Parametric Procedure: Binomial procedure

- Graphic: Binomial plot (`binom.plot(x, n)`)
- Requires: Data generated from Binomial distribution
- `R` function: `binom.test(x, n)`

Note: This is *not* the procedure Hawkes covers. They use something close to what we refer to as the Wald test. The function to perform the Wald test in `R` is `wald.test`.

One-Parameter Procedures: p

Non-Parametric Procedure: Non-parametric bootstrap

- Graphic: Binomial plot (`binom.plot(x, n)`)
- Requires: Nothing
- `R` code:

```
st = numeric()

for (i in 1:1e4) {
    x = sample(y, replace=TRUE)
    st[i] = mean(x)
}

quantile(st, c(0.025, 0.975))
```

This code assumes the data, `y`, consist of 0 and 1 values.

One-Parameter Procedures: σ^2

Parametric Procedure: Chi-Square (χ^2) procedure

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Data generated from Normal process
  - Requirement test: Shapiro-Wilk test
    - `R` function: `shapiroTest(x)`
- `R` function: `onevar.test(x)`

One-Parameter Procedures: σ^2

Non-Parametric Procedure: Non-parametric bootstrap

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Nothing
- `R` code:

```
st = numeric()

for (i in 1:1e4) {
    x = sample(y, replace=TRUE)
    st[i] = var(x)
}

quantile(st, c(0.025, 0.975))
```

Two-Parameter Procedures: $\mu_1 - \mu_2$ (Independent)

Parametric Procedure: two-sample t-procedure

- Graphic: side-by-side box-and-whiskers plot (`boxplot(x ~ g)` or `boxplot(x1, x2)`)
- Requires: Data generated from Normal distribution – in *each* population
  - Requirement test: Shapiro-Wilk test
    - `R` function: `shapiroTest(x ~ g)`
- `R` function: `t.test(x ~ g)`
- `R` function: `t.test(x1, x2)`

Note: This is not the same test as Hawkes uses.

Two-Parameter Procedures: $\mu_1 - \mu_2$ (Independent)

Non-Parametric Procedure: Mann-Whitney procedure

- Graphic: side-by-side box-and-whiskers plot (`boxplot(x ~ g)` or `boxplot(x1, x2)`)
- Requires: Beyond the scope of this course
- `R` function: `wilcox.test(x ~ g, conf.int=TRUE)`
- `R` function: `wilcox.test(x1, x2, conf.int=TRUE)`

Two-Parameter Procedures: $\mu_1 - \mu_2$ (Dependent)

Parametric Procedure: two-sample dependent procedure

- This is just a one-sample procedure if you focus on the differences, $d_i = x_{1i} - x_{2i}$ (whose mean is $\mu_d = \mu_1 - \mu_2$), instead of the measurements.
- As such, refer to the one-sample mean procedure.
- This should help to illustrate the need to understand the data and what you want to understand.

Two-Parameter Procedures: $p_1 - p_2$

Parametric Procedure: Proportions procedure

- Graphic: Binomial plot (`binom.plot(x=c(x1, x2), n=c(n1, n2))`)
- Requires: Expected numbers of successes and failures are at least 5 in each group
- `R` function: `prop.test(x=c(x1, x2), n=c(n1, n2))`

Note: This is *not* the procedure Hawkes covers. They use something close to this, but this procedure makes adjustments for the fact that the Binomial distribution is discrete and the Normal distribution is not. As such, you should use the Wald test for this type of Hawkes problem, too. (`R` function: `wald.test`)

Two-Parameter Procedures: $\frac{\sigma_1^2}{\sigma_2^2}$

Parametric Procedure: Fisher's F-test

- Graphic: side-by-side box-and-whiskers plot (`boxplot(x ~ g)` or `boxplot(x1, x2)`)
- Requires: Data generated from Normal distribution – in *each* population
  - Requirement test: Shapiro-Wilk test
    - `R` function: `shapiroTest(x ~ g)`
- `R` function: `var.test(x ~ g)`
- `R` function: `var.test(x1, x2)`